Kellen Heraty

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Education

University of Washington Expected June 2030
Computer Science, undergraduate *Seattle, WA*

Lakeside School Graduated June 2026
High school diploma · GPA: 3.98/4.0

Research and publication

Lead Researcher Mar 2025–Present
Sculptor AI, independent research organization *Seattle, WA*

- Developed neural posterior estimators for brown dwarf atmospheres using ExoJAX simulations and normalizing flows; generated two matched million-spectrum datasets on Google Cloud TPUs.
- Improved nominal 90% carbon-to-oxygen interval coverage from 61.4% to 89.4% on 2,048 perturbed simulated spectra using opacity marginalization and matched three-seed ensembles.
- Applied the method to 105 JWST spectral products of 13 brown dwarfs; measured 1.3-second, 400-sample posterior inference for one network on a single CPU core.
- Released research code, model weights, and evaluation datasets; also developed the Ursa Minor language model and Constellation platform for access to multiple AI providers.

Kellen Heraty. *Amortized Opacity Marginalization Improves C/O Interval Calibration for Brown-Dwarf Retrievals*. arXiv:2609.01665, September 2026. Preprint.

Research code · Data and models

Experience

Software Engineering Intern Jun–Oct 2024
Microsoft, Global Secure Access *Remote*

- Built a migration tool from Netskope to Microsoft Entra Private Access using Microsoft Graph and Netskope APIs.
- Developed its React and Fluent UI web interface and wrote documentation and demonstration materials.

Information Technology Summer Intern Jun 2023–Aug 2025
Lakeside School *Seattle, WA*

- Led development and deployment of Andromeda AI, a school-wide platform serving hundreds of students with access to multiple AI providers.
- Developed tools for tutoring and code evaluation; performed code reviews and supported school hardware and network upgrades.

Selected projects, awards, and speaking

Navigate on Chestnut September 2026
3rd place, COMMA_HACK 7 · Demo

- Restored Navigate on openpilot with Chase Culbertson, conditioning the driving model on a navigation map to follow turns.
- Demonstrated the system in BeamNG.drive with comma four and chestnut hardware in the loop running model inference.

TEDx LakesideSchool Youth March 2026
Delivered a talk on AI safety and recent developments in artificial intelligence.

Technical skills

Languages: Python, JavaScript, C++, C#, GDScript
ML frameworks and compute: PyTorch, JAX, Hugging Face Transformers, Google Cloud TPUs
ML methods: Model training and fine-tuning, model evaluation, uncertainty calibration, normalizing flows
Software: React, Node.js, Git, Docker, REST APIs, PostgreSQL, Godot